

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution																				
(a)	Obtains one missing expected value eg Low and Breed A = $\frac{21 \times 27}{105}$	AO1.1a	M1	<table><tr><th></th><th>Low</th><th>Med</th><th>High</th></tr><tr><td>A</td><td>5.4</td><td>9.6</td><td>6</td></tr><tr><td>B</td><td>5.14</td><td>9.14</td><td>5.71</td></tr><tr><td>C</td><td>8.23</td><td>14.63</td><td>9.14</td></tr><tr><td>D</td><td>8.23</td><td>14.63</td><td>9.14</td></tr></table>		Low	Med	High	A	5.4	9.6	6	B	5.14	9.14	5.71	C	8.23	14.63	9.14	D	8.23	14.63	9.14
		Low	Med		High																			
A	5.4	9.6	6																					
B	5.14	9.14	5.71																					
C	8.23	14.63	9.14																					
D	8.23	14.63	9.14																					
	Obtains all expected values. A0 if integers	AO1.1b	A1																					
(b) (i)	States both hypotheses using correct language	AO2.5	B1	H_0 Milk yield is independent of breed																				
	States critical value (or p -value)	AO1.1b	B1	H_1 Milk yield is not independent of breed																				
	Evaluates the χ^2 test statistic by comparing the cv with the ts (or p -value with 0.01)	AO3.5a	R1	1 tail 1% $ts = \sum \frac{(O - E)^2}{E} = 19.4$ χ^2 cv for 6 df = 16.81																				
	Infers H_0 rejected	AO2.2b	E1	($p = 0.00275$)																				
	Concludes correctly in context (conclusion should not be definite)	AO3.2a	E1	$219.4 > 16.81$ ($0.00275 < 0.01$) Reject H_0 and conclude that there is evidence to suggest that milk yield is not independent of breed																				
(ii)	Considers $\frac{(O - E)^2}{E}$ or $(O - E)$ to identify largest sources of association [Do not allow mark if no reference to why source selected]	AO2.4	E1	Largest sources of association Breed A/High and Breed B/ Low.																				
	Interprets main source(s) of association in context	AO3.2a	E1dep	Far more than expected Breed A cows observed to have high milk yield or Far more than expected Breed B cows observed to have low milk yield																				

Q2.

(a)

Expected values				
	F	T	S	D
<3	8.736	34.944	24.192	16.128
>3	4.264	17.056	11.808	7.872

Any two correct to 2 d.p.

M1

All correct, here or below, to 2 d.p.

A1

Must be **expected** value, applied to this case, not just general statement.

E1

One expected value for Flat < 5
So combine first two columns to give

Expected values			
	F+T	S	D
<3	43.68	24.192	16.128
>3	21.32	11.808	7.872

For combining first two E columns,
at least 1 correct.

B1

H_0 : No association between property type and time to sell.

H_0 seen somewhere in solution

H_1 : Association between property type and time to sell.

If "independent" used then must be correct way round

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
38	43.68	0.7386
27	21.32	1.5132
28	24.192	0.5994
8	11.808	1.2281
18	16.128	0.2173
6	7.872	0.4452
	χ^2	4.7418

For attempt at $(O_i - E_i)^2 / E_i$

$$4.7 < \chi^2 < 4.8$$

M1

A1

2 degrees of freedom

PI by 4.605, 5.991, 7.378, 9.210 or 10.597 seen

B1

C.V. of χ^2 for 2 d.f. = 4.605

AWFW 4.60 to 4.61

B1

4.74 > 4.605 so reject H_0
 significant evidence of an association
 between property type and time to sell.

Context conclusion.

*Dep. on B1 for H_0 , A1 for χ^2 and
 B1 for c.v.*

A1

10

- (b) (i) More in total than any other type so
 likely to have biggest effect
Or similar referring to large number

E1

- (ii) Far away from expected values
Or opposite pattern to other three

E1

2

If Flats and Detached combined:

Expected values			
	F+D	T	S
<3	24.864	34.944	24.192
>3	12.136	17.056	11.808

For combined F and D

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
22	24.864	0.3299
34	34.944	0.0255
28	24.192	0.5994
15	12.136	0.6759
18	17.056	0.0522
8	11.808	1.2281
	χ^2	2.9110

For attempt at $(O_i - E_i)^2 / E_i$

©2025 Exam Papers Practice. All rights reserved. M1

A0

Accept H^0

A0

Max of 8 marks

2

Q3.

(a) H_0 : No association between A level grade and class of degree

H_1 : Association between A level grade and class of degree

At least H_0 correct

B1

O_i	E_i
20	11.6

9	17.4
36	36.4
55	54.6
22	28
48	42
2	4
8	6
200	200

For E_i 's attempted

M1

Combine Class 2(ii) and 3

For combining attempted

M1

20	11.6	8.4	6.0827
9	17.4	-8.4	4.0552
36	36.4	-0.4	0.0044
55	54.6	0.4	0.0029
24	32	-8	2.0
56	48	8	1.3333
200	200	0	13.47

For final column attempted

M1

(awrt 13.5)

A1

$v = 2$

[$v = 3$ with $\chi^2 = 11.345$ (B0B1ft)]

B1

©2025 Exam Papers Practice. All rights reserved.

$\chi^2_{1\%}(2) = 9.210$

B1

Reject H_0

Dep on B1 M1M1M1 B1B1, not A1

A1

Fiona's belief justified

Dep on B1 M1M1M1 B1B1, not A1

E1

9

- (b) **Fewer than expected gained a Class 1**
degree having gained grade B in A-level Mathematics.

E1

**More than expected gained a Class 2(ii)**

degree having gained grade B in A-level Mathematics.

E1

correct comments (see below)

1	2(i)	2(ii)	3	comb
9	55	48	8	56
17.6	54.6	42	6	48
A	B	C	D	E

*A: fewer than expected**B: as expected**C: more than expected**D: more or similar than expected**E: more than expected*

2

[11]

Q4.

- (a) H_0 : no association (between type of school and performance of 16 year olds in their GCSEs)

H_0 : type of school and performance of 16 year olds in their GCSEs independent

B1

1

(b)
$$\frac{(O - E)^2}{E}$$

0.195819311

0.482160711

0.003569447

1.080536181

0.062507172

1.269422099

0.785491128

0.183802623

0.541856652

0.044011976

3.274102564

4.096492891

'Attempt at
$$\frac{(O - E)^2}{E}$$

(≥ 4 correct values seen to 2dp)

M1

$$X^2 = \sum \frac{(O - E)^2}{E}$$

Attempt to add ≥ 8 terms

m1

$$= 12.01977275$$

$$= 12.0 \text{ (1dp)}$$

$$\text{Allow } 11.9 \leq X^2 \leq 12.1 \Rightarrow M1m1$$

CAO

A1

3

(c) $v = 6 \Rightarrow \chi^2_{1\%} = 16.8(12)$

$$v = 6 \text{ can be implied by } \chi^2_{1\%} = 16.8(12)$$

B1,B1

No (significant evidence to suggest an) **association between**
(type of) **school and (GCSE) performance** (of 16 year olds)

Insufficient/no evidence to support Emily's belief.

School and performance are independent.

*Correct conclusion **in context***

*Dep on B1M1m1B1B1 given in (a), (b), (c) **and***

$$11.9 \leq X^2 \leq 12.1$$

Adep1

3

- (d) More than expected gained at least / more than 5 GCSEs
Fewer than expected gained at least / more than 1 GCSE
but less than 5 GCSEs
Fewer than expected gained no GCSEs

*Since conclusion of **no association** between school
and GCSE performance, it may be misleading to look at
individual differences in any great detail*

Any one of these 4 comments seen

©2025 Exam Papers Practice. All rights reserved.

B1

1

(e) $\chi^2_{10\%} = 10.6(45)$

Correct value of χ^2 only

B1

Reject H_0 at 10% level of significance.

(Evidence to suggest) an **association between** (type of)
school and (GCSE) performance

Evidence to support Emily's (Joanne's) belief.

(Type of) school + (GCSE) performance dependent.

*Dep on B1M1m1 **and** $11.9 \leq X^2 \leq 12.1$ **and** B1 in (e)*

Bdep1

2

[10]

Q5.

- (a) H_0 : no association between age and first time performance in driving test

H_1 : association between age and first time performance in driving test

B1

	Pass		Fail		
Age	O	E	O	E	Total
17-18	28	19.2	20	28.8	48
19-30	2	6.4	14	9.6	16
31-39	12	18.0	33	27.0	45
40-60	6	4.4	5	6.6	11
Total	48	48	72	72	120

E's attempted

Correctly

M1

A1

O	E	$(O - E)^2 / E$
28	19.20	4.0333
2	6.40	3.0250
18	22.40	0.8643
20	28.80	2.6889
14	9.6	2.0167
38	33.6	0.5762
		13.20

Attempt at combining

Correctly

M1

A1

Final column attempted

m1

For X^2 correct

A1

$$v = 2 \Rightarrow \chi^2(2) = 9.210$$

(on $v = 2$ or $v = 3$ only)

B1ft

Reject H_0
Evidence to support Julie's belief
at 1% level of significance.

E1ft

9

- (b) More students than expected in the age group 17-18 pass their test first time.

Fewer than expected fail

E1

1

[10]

Q6.

(a)

O_i	E_i	$ O_i - E_i - 0.5$	$\frac{7.5^2}{E_i}$
52	44	7.5	1.2784
58	66	7.5	0.8523
28	36	7.5	1.5625
62	54	7.5	1.0417
			4.7349

E attempted

M1

Yates' correction attempted

M1

χ^2 attempted

M1

AWFW 4.73 to 4.74

A1

H_0 : No association between incidence
of asthma and volume of traffic

H_1 : Association

at least H_0 stated correctly

B1

$$v = 1$$

$$\chi^2_{\text{crit}} = 3.841 < 4.7349$$

critical value

B1

Reject H_0 at 5% level

A1ft

Evidence to suggest an association
between the incidence of asthma in
children and the volume of traffic where they live

E1ft

8

- (b) More than expected had asthma
dep on statement of association

E1

1

[9]

Q7.

- (a) H_0 : No association between the
performances at KS3 and GCE

B1

O_i	E_i	$O_i - E_i$	X^2
60	63.55	-3.55	0.1983
55	44.64	10.36	2.4043
40	46.81	-6.81	0.9907
55	51.25	3.75	0.2744
32	36.00	-4.00	0.4444
38	37.75	0.25	0.0017
47	46.33	0.67	0.0097
31	32.54	-1.54	0.0733
35	34.13	0.87	0.0222
43	43.87	-0.87	0.0173
26	30.82	-4.82	0.7527
38	32.31	5.69	1.0005

E_i M1

$O_i - E_i$ M1

$(O_i - E_i)^2 / E_i$

Σ M1

$$X^2 = 6.1897$$

AWFW 6.05 – 6.35

A1

$$\nu = 3 \times 2 = 6 \Rightarrow \chi^2_{90\%} = 10.645$$

on their ν

B1B1ft

Do not reject H_0
 No evidence to suggest an association
 between KS3 results and GCE grades at
 10% level of significance.

E1ft

9

- (b) More of the students achieving level 7 at
 KS3 gain grade A's at GCE than expected.

E1

1

[10]

Q8.

- (a) H_0 : Choice independent of gender
gender not associated with choice

B1

	Squash	Badminton	Archery	Hockey
Male	5/3.5	16/14	30/24.5	19/28
Female	4/5.5	20/22	33/38.5	53/44

M1

Combine Squash and Badminton

$E_i < 5$ (Similar categories)

M1

	S & B	Archery	Hockey	
Male	21/17.5	30/24.5	19/28	M1
Female	24/27.5	33/38.5	53/44	M1

χ^2 values

	S & B	Archery	Hockey	
Male	0.7000	1.2347	2.8928	M1
Female	0.4455	0.7857	1.8409	

$\chi^2_{\text{calc}} = 7.90$

(7.8 to 7.9)

A1

$$v = 2$$

B1

$$\chi^2_{5\%}(2) = 5.991$$

(on their v)

B1ft

10

Reject H_0 .

Sufficient evidence, at the 5% level of significance, to support an association between the choice of sport and gender

reject H_0 and H_0 stated

or

statement in context

(b) **More females and fewer males** chose

B1

to participate in hockey **than expected**

B1

2

[12]

Q9.

(a) (i)

	A	B	Total
22-34	21	32	53
35-39	72	36	108
40-59	27	12	39
Total	120	80	200

for A values

B1

for B values

B1

2

(ii) H_0 : no association between area and age profile.

H_1 : association between area and age profile.

B1

At least H_0

O_i	E_i	
21	31.8	3.6679
72	64.8	0.8000
27	23.4	0.5538
32	21.2	5.5019
36	43.2	1.2000
12	15.6	0.8308
$\sum O_i = 200$	$\sum E_i = 200$	$\chi^2 = 12.554$

attempt row & column totals

M1

attempt at E_i

M1

attempt at $\frac{(O_i - E_i)^2}{E_i}$

M1

attempt at χ^2

M1

(AWFW 12.5 to 12.6)
provided correct method used

A1

©2025 Exam Papers Practice. All rights reserved.

$$v = (3 - 1)(2 - 1) = 2$$

B1

$$\chi^2_{1\%}(2) = 9.210 < 12.554$$

on their v and χ^2

B1ft

Reject H_0

The evidence suggests that the area within which a school is situated seems to have an effect on the age-profile of the staff employed.

on χ^2 and calculated value dep on H_0 correct, if stated.

E1ft

- (b) There seems to be fewer staff employed in 22 - 34 age group than expected in school A and more than expected in school B

E1

E1

2

[13]

Q10.

(a)

	D	P	DN
< 65	4 2.70	12 7.72	6 11.58
≥ 65	10 11.30	28 32.28	54 48.42

Method for E's

M1

Pooling D&P

	D&P	DN
< 65	16 10.42	6 11.58
≥ 65	38 43.58	54 48.42

correct method for pooling - must be D&P

m1

H_0 : Answer not associated with age

B1

H_1 : Answer associated with age

correct null hypothesis - may be implied by clearly stated conclusion

$$\frac{(|O-E| - 0.5)^2}{E} =$$

attempt at $\sum \frac{(O-E)^2}{E}$ - their figures

M1

attempt at Yates' correction

m1

correct method for Yates'

m1

$$\frac{5.08^2}{\left(\frac{1}{10.42} + \frac{1}{11.58} + \frac{1}{43.58} + \frac{1}{48.42}\right)} = 5.83$$

5.83 (5.81 to 5.85)

A1

c.v χ^2_1 is 3.841

1df or 2df if no pooling

B1

3.841 or 3.84 (allow 5.991/5.99 if no pooling)

B1

Reject H_0 , conclude evidence to suggest that answer is associated with age.

correct conclusion - must be compared with upper tail of χ^2

A1ft

10

- (b) Older delegates more likely to answer that they would not change their votes.

Older delegates less likely to answer that they would change their votes

E1

1

- (c) Most delegates are over 65

Most delegates are elderly

E1

1

[12]

Q11.

(a)

b		CC		IC		S
<120	2	16.94	58	32.94	20	30.12
≥ 120	34	19.06	12	37.06	44	33.88

Method for one E

M1 m1 B1

Method for all E's

M1 m1

H_0 : No association between asking price and area
Correct H_0 - may be implied by conclusion

B1

H_1 : Asking price associated with area

$$\sum \frac{(O - E)^2}{E} = 67.3$$

Attempt at $\sum \frac{(O - E)^2}{E} = 67.3$

Completely correct method

67.3(67, 68) - allow if demonstrated to be greater than c.v.

M1 m1 A1

Critical value χ^2_{2df} for 1% risk 9.210
ft 2df

B1ft

Reject H_0 , conclude asking price
ft c.v. - candidate's df

B1ft

associated with area.

ft correct conclusion - must be compared with upper tail of χ^2 (requires M1M1 only)

©2025 Exam Papers Practice. All rights reserved.

A1ft

9

- (b) Houses appear to be cheapest/proportion costing < £120 000 greatest, in inner city followed by suburban areas and most expensive in city centre

Cheapest in inner city

Most expensive in city centre

E1 E1

2

[11]

Q12.

(a)	12 – 3	3 – 6	6 – 9
12	22.58	6	12.51
54		35	28.49
62	51.42	67	84.08

Method for one E

M1

Method for all E's

M1

H_0 : Alcohol independent of time of day

Correct null hypothesis - may be earned in conclusion

B1

H_1 : Alcohol not independent of time of day

$$\sum \frac{(O - E)^2}{E} = 23.4$$

Any attempt at $\sum \frac{(O - E)^2}{E}$

M1

Completely correct method

m1

23.4 (23.2 – 23.6)

A1

cv χ^2 for 1% 1 - sided risk is 9.210
2d.f.

B1

9.21 - upper 1% point of χ^2 , their d.f.

B1 ft.

©2025 Exam Papers Practice. All rights reserved.

Reject H_0 , conclude taking alcohol with meal associated with time of day

Conclusion, must use upper tail of χ^2 , depends on all method marks

A1 ft.

9

- (b) (i) A greater proportion of customers entering between 6 pm and 9 pm bring alcohol than those entering at other times
Clear interpretation gets both marks

E2(1)

- (ii) Customers enter at a greater rate between 6 pm and 9 pm than at other times
Customers enter at greater rate

E1

3

- (c) Ideally a random sample of all customers should be included but this is impractical.
Conclusions only invalid if chosen day

E1

was not representative of all days e.g. if pattern was different at weekends.

Might affect conclusion

Reason

E1

2

[14]

Q13.

- (a) H_0 : Reason not associated with time of day

Correct null hypothesis – may be implied by clearly stated conclusion

B1

H_1 : Reason associated with time of day

	Rush hour		Out of RH	
Normal	34	27.05	45	51.95

Method for one E

M1

	Beatles		28	
	4	10.95	28	21.05

Method for all E's

m1

$$\frac{\sum (O - E)^2}{E} =$$

$$\frac{\sum (O - E)^2}{E}$$

Attempt

Attempt Yates's correction

Clearly correct method for Yates's

M1 m1 m1

$$6.45^2 \left(\frac{1}{27.05} + \frac{1}{51.95} + \frac{1}{10.95} + \frac{1}{21.05} \right) = 8.11$$

$$8.11 (8.05 - 8.15)$$

A1

c.v. χ^2 is 3.841

1 d.f.

3.841 or 3.84

B1 B1ft

Reject H_0 conclude reason associated with time of day

Conclusion – t.s. must be compared with upper tail of χ^2 - needs first 3 M marks

A1ft

10

- (b) Of those using the crossing because of the Beatles, a greater proportion do so out of rush hour than during rush hour

*Greater **proportion** out of rush hour*

E1

1

[11]

Q14.

(a)

	J		R		S	
F	16	10.77	12	10.00	22	29.23
M	12	17.23	14	16.00	54	46.77
	28		26		76	130

for at least one E

for all E's

©2025 Exam Papers Practice. All rights reserved.

M1 M1

H_0 : Choice of language independent of gender

correct null hypothesis (may be implied by clearly stated conclusion)

B1

H_1 : Choice of language not independent of gender

$$\sum (O - E)^2/E = 5.23^2/10.77 + 5.23^2/17.23 + 2^2/10 + 2^2/16 + 7.23^2/29.23 + 7.23^2/46.77 = 7.68$$

attempt at $\sum (O - E)^2/E$ using their E's - must be for all cells 7.6 ~ 7.7

M2 A1

c.v. χ^2 for 5% significance level is 5.991



2 degrees of freedom correct c.v.,
their degrees of freedom

B1 B1ft

Reject H_0 , Conclude choice of language associated with gender.
*correct conclusion their c.v. - test statistic must be compared
with upper tail of χ^2 - needs all previous M marks.*

A2ft

10

- (b) (i) Swedish more popular with males than with females

B1

1

- (ii) Swedish is most popular choice among both males and females
both marks for clearly expressed interpretation.

B2

2

[13]

Q15.

- (a) (i) H_0 : Irritation not associated with carrying phone
*Correct null hypothesis - if not stated may be implied
by final correct conclusion*

H_1 : Irritation associated with carrying phone
*Needs clear statement; can be earned in conclusion
or (a)(ii)*

	Irritated		Not irritated	
Phone	12	17.03	14	8.97
No phone	26	20.97	6	11.03

Correct method for one E

B1

Correct method for all E's

M1 M1

$$\sum \frac{(|O - E| - 0.5)^2}{E}$$

Attempt
$$\sum \frac{(O - E)^2}{E}$$

Attempt at Yates's correction

M1 m1

$$= 4.53^2 \left(\frac{1}{17.03} + \frac{1}{8.97} + \frac{1}{20.97} + \frac{1}{11.03} \right) = 6.33$$

*Correct method for Yates's
(6.2 - 6.5) or demonstration*

$$\sum \frac{(O - E)^2}{E} \text{ exceeds their c.v.}$$

m1 A1

$$\text{cv } \chi^2_{1} 5\% = 3.841$$

*for 1 degree of freedom
3.841 or 3.84*

B1 B1

Reject H_0 : irritation associated with carrying mobile phone.

*Correct conclusion – requires first M1m1M1 and
upper tail of χ^2 used*

A1ft

10

- (ii) Passengers not carrying mobile phones more likely to be irritated than those carrying mobile phones

Conclusion in context

E1

1

- (b) (i) Combining dissimilar classes makes interpretation difficult or impossible

Interpretation difficult

E1

1

- (ii) Not necessary to combine extremely irritated as $E > 5$

Not necessary $E > 5$

E1

Anthony's suggestion leaves 3×2 table to be analysed which is more likely to establish an association if one exists (more degrees of freedom).

More likely to establish a difference / more degrees of freedom

E1

2

- (iii) H_0 : Irritation not associated with carrying phone

H_1 : Irritation associated with carrying phone

Null hypothesis - if not stated may be implied by final correct conclusion

B1

	Not / Slightly		Irritated		Extremely	
Phone	52	39.70	18	20.49	3	12.81
No Phone	10	22.30	14	11.51	17	7.19

$$\sum \frac{(O - E)^2}{E}$$

Attempt at $\sum \frac{(O - E)^2}{E}$

M1 M1

$$= \frac{12.30^2}{39.70} + \frac{12.30^2}{22.30} + \frac{2.49^2}{20.49} + \frac{2.49^2}{11.51} + \frac{9.81^2}{12.81} + \frac{9.81^2}{7.19} = 32.3$$

(32.2 - 32.5) or demonstration

$$\sum \frac{(O - E)^2}{E} \text{ exceeds their cv}$$

A1

cv $\chi^2_{2, 1\%} = 9.210$

2 degrees of freedom
9.210 or 9.21

B1 B1

Reject H_0 , irritation associated with carrying phone.

Correct conclusion - requires previous M1 only and upper tail of χ^2 used.

A1ft

Passengers not carrying mobile phones more irritated than others

Conclusion in context

E1

8

[22]

Examiner reports

Q2.

Most students followed the instruction to combine two columns, though some did not explain why this was necessary, and others ignored it completely. Only a small proportion thought it was appropriate to combine 'flats' with 'detached'; the vast majority making the obvious choice of 'flats' and 'terraced'. The calculations were generally well done, although some did not work with sufficient significant places. Only a few felt it necessary to use Yates' correction. The majority of students realised that the degrees of freedom had now been reduced to 2 and reached the correct conclusion. In the final part, when students chose difference between expected and observed values as their criterion, it was often unclear why 'terraced' might be chosen in preference to 'flats'. There were, however, many excellent explanations.

Q3.

As has been the case on previous papers, this was again one of the best answered questions. Most students managed to state correct hypotheses and then evaluate the required expected values. Most realised that 'degree classes 2(ii) and 3' needed to be combined and did so successfully, and for the correct reason that $E_i < 5$ for those students gaining a 'degree class 3' degree having gained an 'A-level grade A'. This usually led to the correct final column and hence the correct value of 13.5 for X^2 . Correct conclusions, in context, were then usually seen. Unfortunately some students, who realised that they had to combine in some way, decided incorrectly, to combine 'degree classes 1 and 3 or 2(i) and 3'. This obviously lost marks. Other students combined 'degree classes 2(i) and 2(ii)' and consequently lost most of the marks for the question whilst others combined for the wrong reason by indicating that $O_i < 5$ necessitated combining cells, or failed to combine any 'degree classes'. There were many excellent answers given to part (b). However, incorrect statements such as "Those receiving a class 1 obtained fewer grade B's at A-level than expected" gained no credit.

Q4.

As usual, the question on this topic was the best answered question on the paper with many fully correct solutions seen. The vast majority of candidates managed to correctly state at least the null hypothesis in part (a). In part (b), a few lost a mark for failing to give their answer to one decimal place, this despite the specific request. However, on the whole, this was very well done, often in conjunction with part (c) where the correct values of $v = 6$ and $\chi^2 = 16.812$ were usually seen, followed by a correct conclusion in context. Those candidates who stated an incorrect null hypothesis in part (a), or failed to find a value for X^2 within the range $11.9 \leq X^2 \leq 12.1$ in part (b), lost this mark. The responses to part (d) were better than usual but there was still room for improvement in this area. In part (e), many candidates found the correct value of $\chi^2 = 10.645$ but again lost the mark for the comment in context for the same reasons as in part (c).

Q5.

This was the second most common source of marks for the majority of candidates, with many fully-correct solutions seen. However, far too many candidates failed to state their hypotheses but still felt justified in drawing some sort of conclusion. The question required candidates to combine two rows. However, some candidates were thrown by the observed value of 2 which did not give an expected value of less than 5. This usually led to incorrect combining and even to the use of Yates' correction when a 2×2 table

resulted. In fact, many candidates did not attempt to combine at all.

In part (b), only the more able candidates gave the correct interpretation that 'More students than expected pass their test first time' or 'Fewer students than expected fail their test first time'. Many candidates appeared not to understand the question and so simply compared age groups.

Q6.

Although this first question provided most candidates with a good start to the paper, there were some who, in part (a), either failed to state any hypotheses or who thought that the null hypothesis should indicate 'an association', rather than 'no association', between the incidence of asthma and the volume of traffic. Also simply stating " H_0 : not associated" and " H_1 : associated" was not really sufficient when a statement in the context of the question was required. Although the great majority of candidates realised that Yates' correction had to be used, there were far too many candidates who could not apply it correctly. A conclusion in context was required: simply stating "Reject H_0 " was not sufficient to gain full marks. Although most candidates formulated a correct conclusion in context, some either used statements which were too positive in nature or made no statement at all. In part (b), the question asked for a comment about the 52 children who had asthma, given that they lived in an area where the volume of traffic was heavy. It did **not** ask, as many candidates thought, for a comparison with the children who lived in an area where the volume of traffic was light. Consequently, the required response "More than expected had asthma" was not always seen. Some candidates, having drawn the wrong conclusion in part (a) that there was 'no association' between the incidence of asthma and the volume of traffic, still felt justified in making this statement.

Q7.

This type of question is always popular and this year it again proved to be a good source of marks. Most candidates correctly stated the null hypothesis and their conclusions were usually correctly stated in context. Many candidates used their calculator to perform all of

the necessary working and simply stated their value for $\sum_i \left(\frac{(O_i - E_i)^2}{E_i} \right)$ and gave the critical value of $\chi^2 = 10.645$ before then stating their conclusion in context. This is a method to be encouraged, especially for those candidates who can use their calculators proficiently. It should be understood, however, that the wrong value stated for

$\sum_i \left(\frac{(O_i - E_i)^2}{E_i} \right)$, together with no numerical calculations seen to indicate that a correct method has been used, will usually result in a loss of many marks. The responses to part (b) were rather mixed, with the correct answer "More of those students achieving level 7 at KS3 gained a grade A at GCE than was expected" often being seen. However, there were many general comments seen that did not relate to this group of students at all.

Q8.

This question produced many disappointing responses. In part (a), most candidates stated correct hypotheses and usually worked out the required values for the E_i . However, there were still some candidates who insisted on rounding these values to the nearest whole number; which is penalised. The calculations of the E_i were usually followed by an incorrect assumption that $\nu = 3$ and consequently the wrong χ^2 value was quoted from the tables. Although many candidates failed to realise that when $E_i < 5$, similar categories must be combined, the majority went on to use the correct method for evaluating χ^2 . The

conclusions drawn were usually correct and often in context. However, some candidates still think, incorrectly, that the rejection of the null hypothesis proves that the alternative hypothesis must be true. Part (b) proved to be beyond all but the very best candidates. The vast majority failed to interpret their results found in part (a). Most candidates simply referred to the original data, usually stating that 'more females than males played hockey' rather than making the required reference to the expected and observed values.

Q9.

Many candidates failed to realise that frequencies and not percentages must be used when constructing a contingency table. Consequently, in part (a)(i), these candidates gained no credit. The majority of candidates were able to demonstrate the required techniques in part (a)(ii), albeit often using the given percentage values. Sensible conclusions in context and related to stated hypotheses were usually seen. In part (b), most candidates failed to answer the question set. Candidates were expected to compare observed and expected values. Instead, most simply made general comments about the 11-24 age group and, as a consequence, gained no credit.

Q10.

As usual candidates were well able to deal with contingency tables although some lost marks by failing to pool. A few pooled without first calculating any expected values, implying that they had based their decision to pool on the observed rather than the expected values.

Q11.

The great majority of candidates answered part (a) successfully. A few incorrectly thought that the small observed value for city centre properties costing less than £120 000 made it necessary to combine classes. In part (b), many merely repeated their conclusion to part (a) instead of observing that properties appear to be most expensive in the city centre and cheapest in the inner city.

Q12.

The performance of the candidates, as a whole, in analysing the contingency table can only be described as excellent. To gain full marks in parts (b)(i) and (b)(ii) it was necessary to distinguish clearly between the *proportion* and the *number* of customers bringing alcoholic drinks.

Q13.

Nearly all candidates were able to make a good attempt at analysing the contingency table. Only a few failed to attempt Yates' correction, although the use of

$\sum \frac{(O - E - 0.5)^2}{E}$ instead of $\sum \frac{(|O - E| - 0.5)^2}{E}$ was not uncommon. However, the interpretation in part (b) was not well done. The expected answer was that the proportion of people using the crossing because of the Beatles was less in rush hour than outside rush hour.

Q14.

The great majority of candidates scored high marks by carrying out the calculations correctly although some then drew the wrong conclusion. Most offered some relevant interpretation but only a minority gained full marks for part (b). A comment made by many

candidates was that more males than females studied Swedish. This is true but, as there were more male than female students, it was hardly surprising. The data and the test result implied that a significantly greater proportion of males than females studied Swedish.

Q15.

Most candidates scored highly on this question. The mechanics of analysing contingency tables, including, where appropriate, combining classes and the use of Yates's correction, seems well understood. Many had difficulty framing appropriate hypotheses. 'H₀ not linked: H₁ linked' is not adequate, although candidates were allowed to redeem themselves by a clearly stated conclusion which enlarged on exactly what was linked. In part (a)(ii), candidates were expected not only to state that irritation and carrying a mobile phone were associated but that (unsurprisingly) passengers not carrying a mobile phone were more likely to be irritated than those carrying a mobile phone.

In parts (b)(i) and (b)(ii), many candidates scored marks by having more or less the right ideas despite some difficulty in expressing them convincingly. In part (b)(i), most candidates thought it was a bad idea to combine opposites while not going on to comment on the difficulty this would cause when trying to interpret the results. In part (b)(ii), most candidates thought a 3×2 table was preferable to a 2×2 table, without really being sure why. Vague reasons such as 'more accurate' were given. More substantial reasons such as 'less likely to lead to a Type 2 error' were much rarer.